Scalar Load: Iscandar vs Buddy

```
Load: Scalar Load (author's shelf, 2026-10-26)
  request sizes ~ Normal(mu 4.5, sigma 1.17),
  rounded and clamped to 1..8 sectors: scalar
  variable use, the allocator's home turf.

Engines: the full Iscandar engine (all modes,
  exact fit) vs the classic binary buddy.
Arena: 32,768 sectors (512 KiB at 16 B/sector).
The same seeded stream drives both engines.

Part A - the sprint (fill to 85% logical, churn,
push; first failed allocation loses):
  buddy    stalled op 5555 (ramp) at 100.00% physical
  iscandar stalled op 12191 (push) at 97.51% physical
  margin 6636 ops to iscandar.

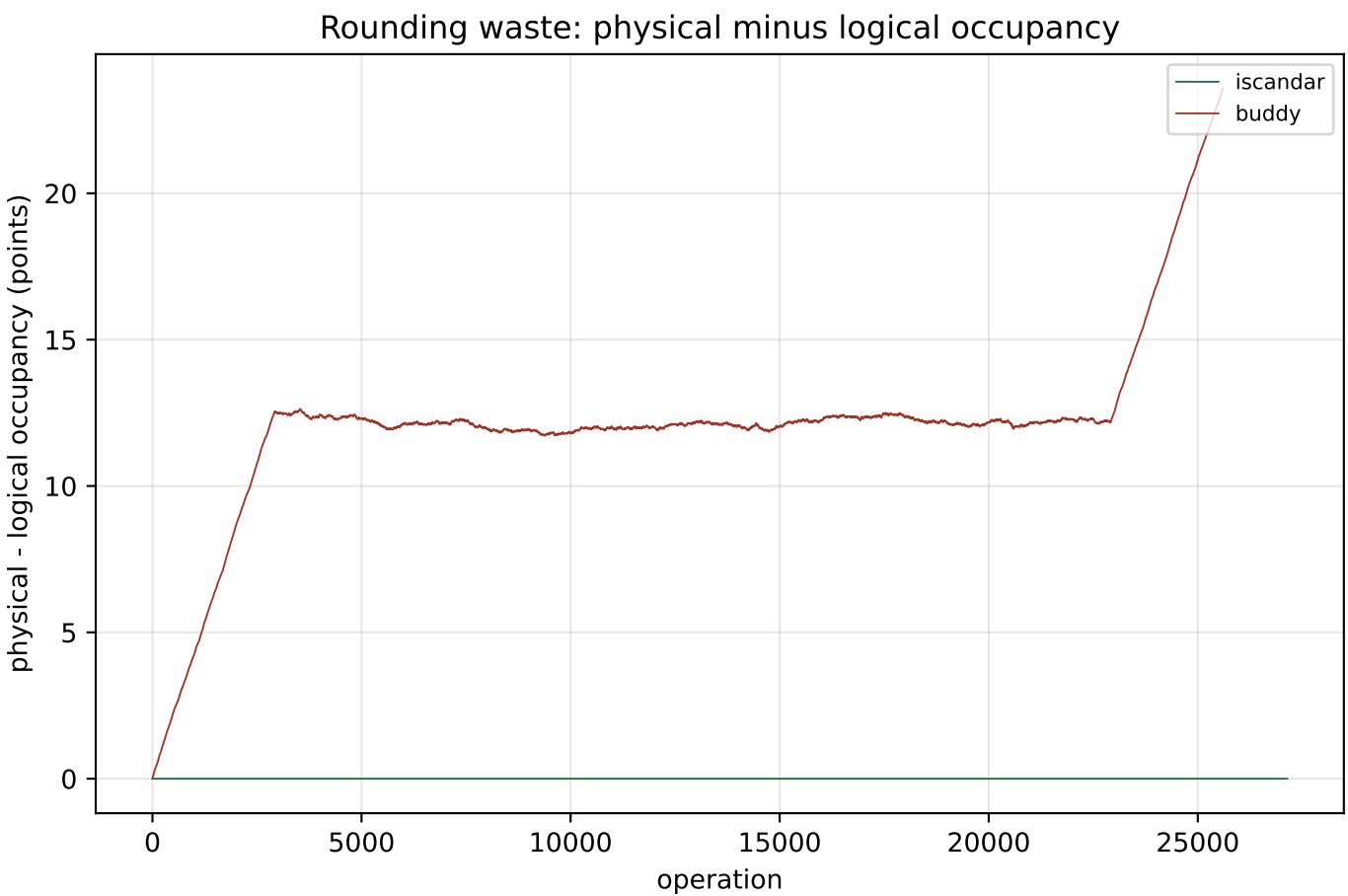
Part B - the steady-state hold (fill to 40%
logical, 20000 ops in a +/-2-point band, push):
  refusals during the hold: iscandar 0, buddy 0
  push stall: buddy at 100.00% physical, largest
            free block 1 sectors
  push stall: iscandar at 97.73% physical, largest
            free run 4 sectors

GammaCheck green in both parts (block checks
every op, full-ladder checks at boundaries).
```


Part B hold: occupancy over time
occupancy (% of cube)
operation
0
20
40
60
80
100
0
5000
10000
15000
20000
25000
iscandar physical
buddy physical
logical (workload)

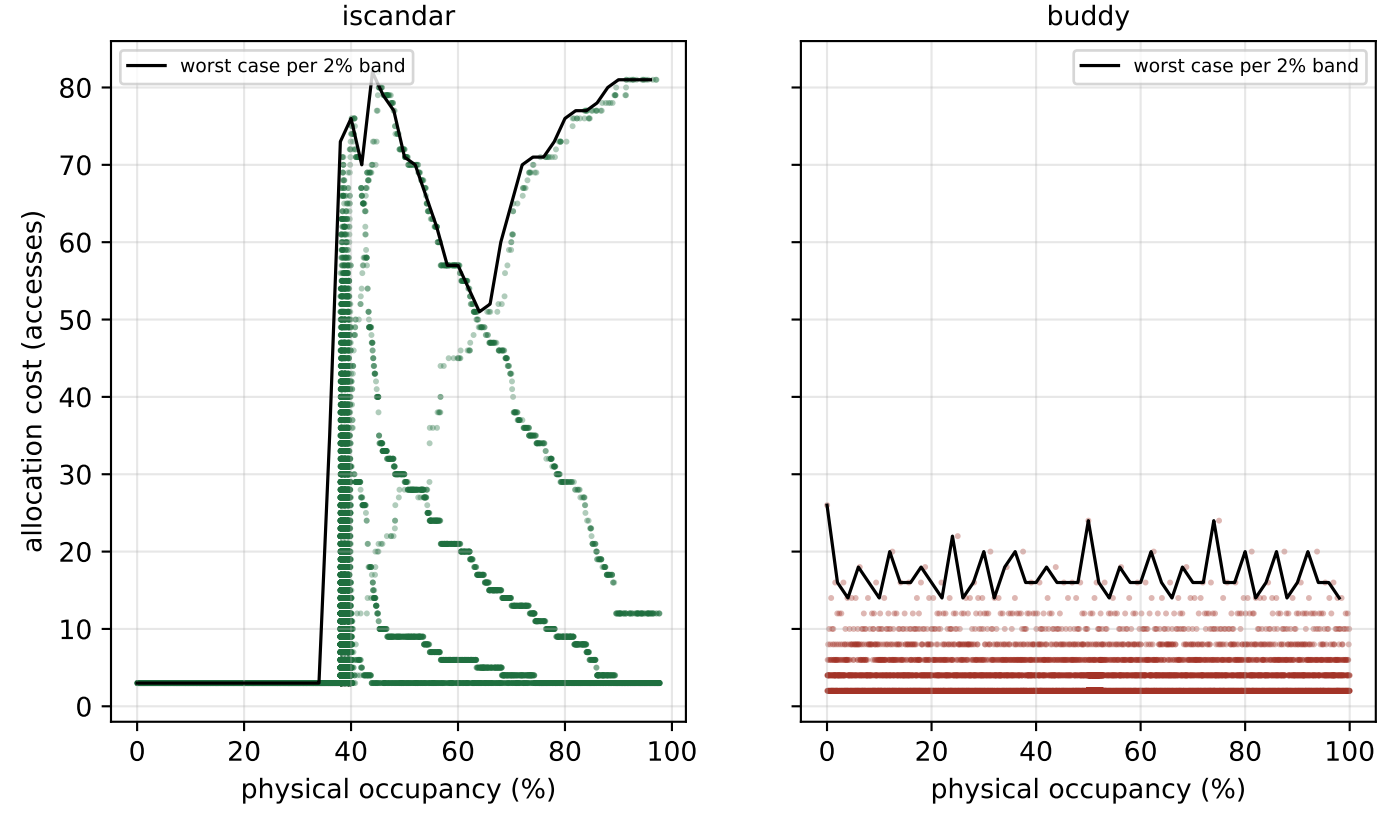
Commentary - occupancy over time

The gray dashed line is the workload itself, held at 40% logical. Iscandar's green line lies on top of it (mean 38.8% physical), because exact fit makes physical occupancy equal to logical occupancy. Buddy's red line rides above it: mean 51.0% physical, ranging 49.7 to 52.9%. The vertical distance between the red line and the workload is space Buddy has consumed without holding data.


Rounding waste: physical minus logical occupancy
iscandar
buddy
physical - logical occupancy (points)
0
5
10
15
20
0
5000
10000
15000
20000
25000
operation

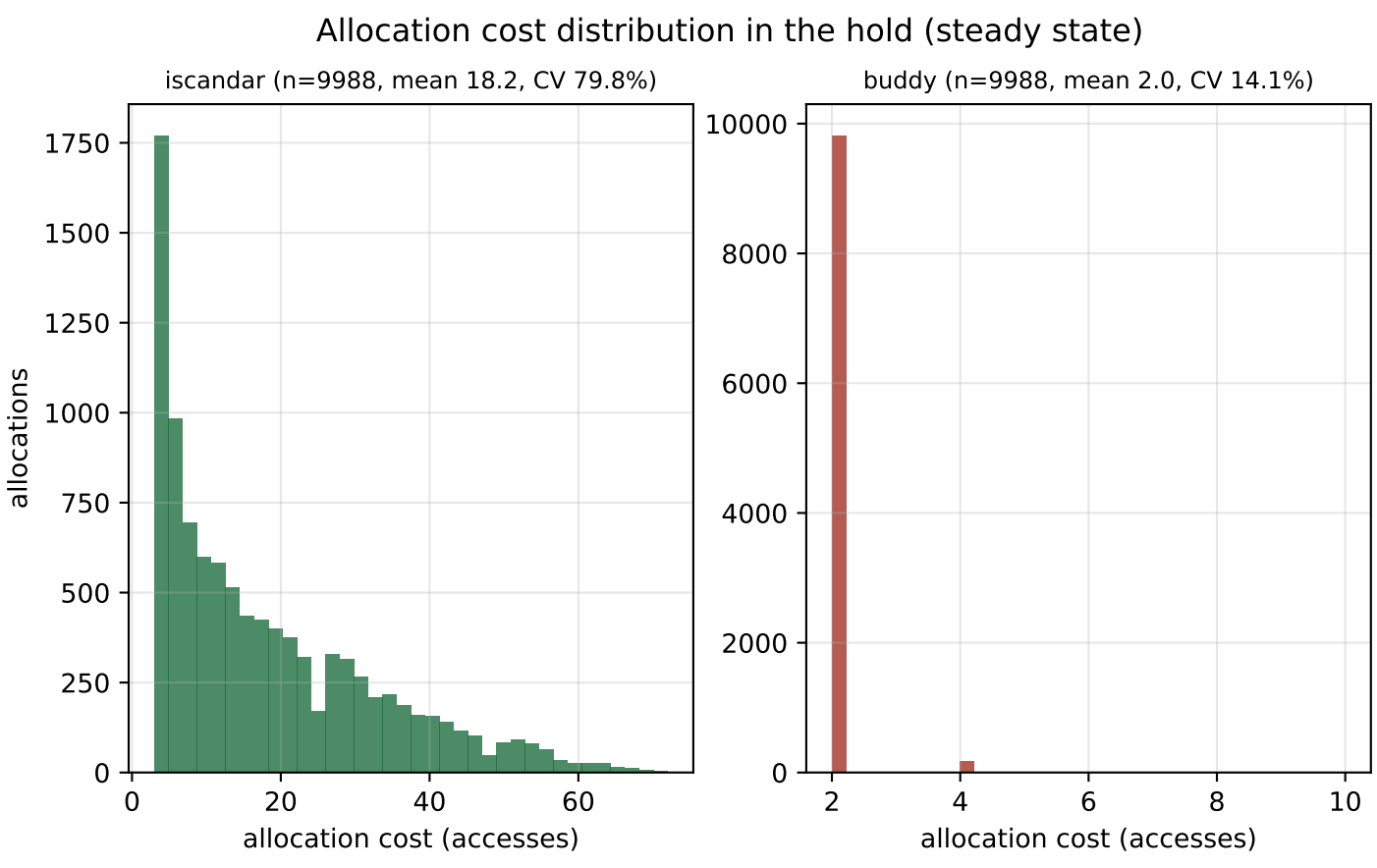
Commentary - rounding waste

Iscandar is a flat line on zero for the entire run. Buddy averages 12.1 points of pure waste and peaks at 12.6 - space that is neither data nor usable, only rounding.


Allocation cost against occupancy (hold + push; units differ per engine, see closing page)
iscandar
buddy
worst case per 2% band
worst case per 2% band
allocation cost (accesses)
physical occupancy (%)
physical occupancy (%)
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100

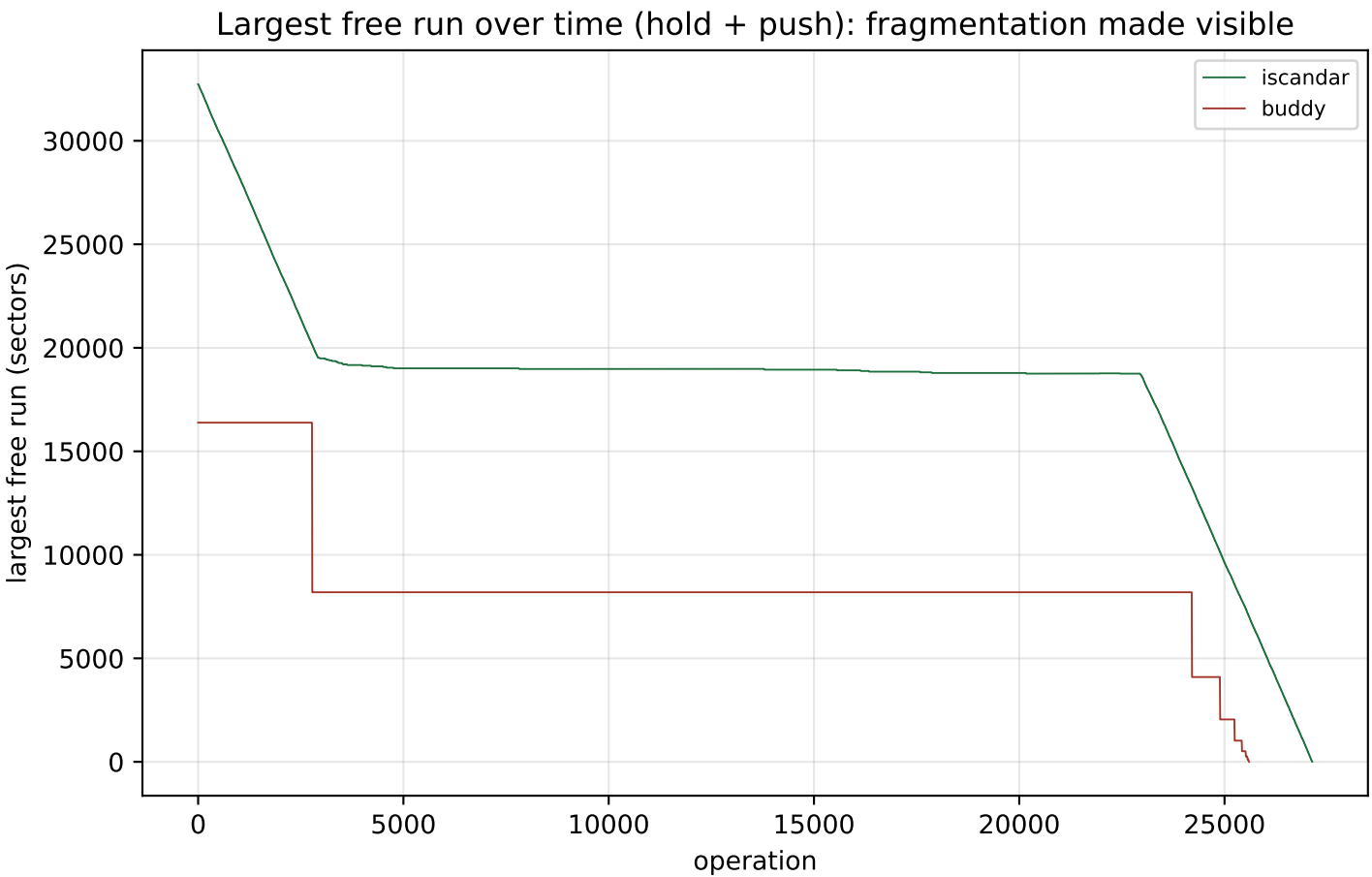
Commentary - cost against occupancy

Two panels with a black worst-case envelope. Iscandar's envelope stays between 3 and 82 touches across every occupancy band it visits; the price does not blow up as the cube fills. Buddy's envelope stays between 14 and 26 in its own tree-visit units (the levels are not comparable across panels; the shapes are). The readable point is the absence of a cliff: neither engine's worst case runs away with occupancy under this load.


Allocation cost distribution in the hold (steady state)
iscandar (n=9988, mean 18.2, CV 79.8%)
buddy (n=9988, mean 2.0, CV 14.1%)
allocations
allocation cost (accesses)
allocation cost (accesses)
0
250
500
750
1000
1250
1500
1750
2000
4000
6000
8000
10000
0
20
40
60
2
4
6
8
10

Commentary - cost distributions

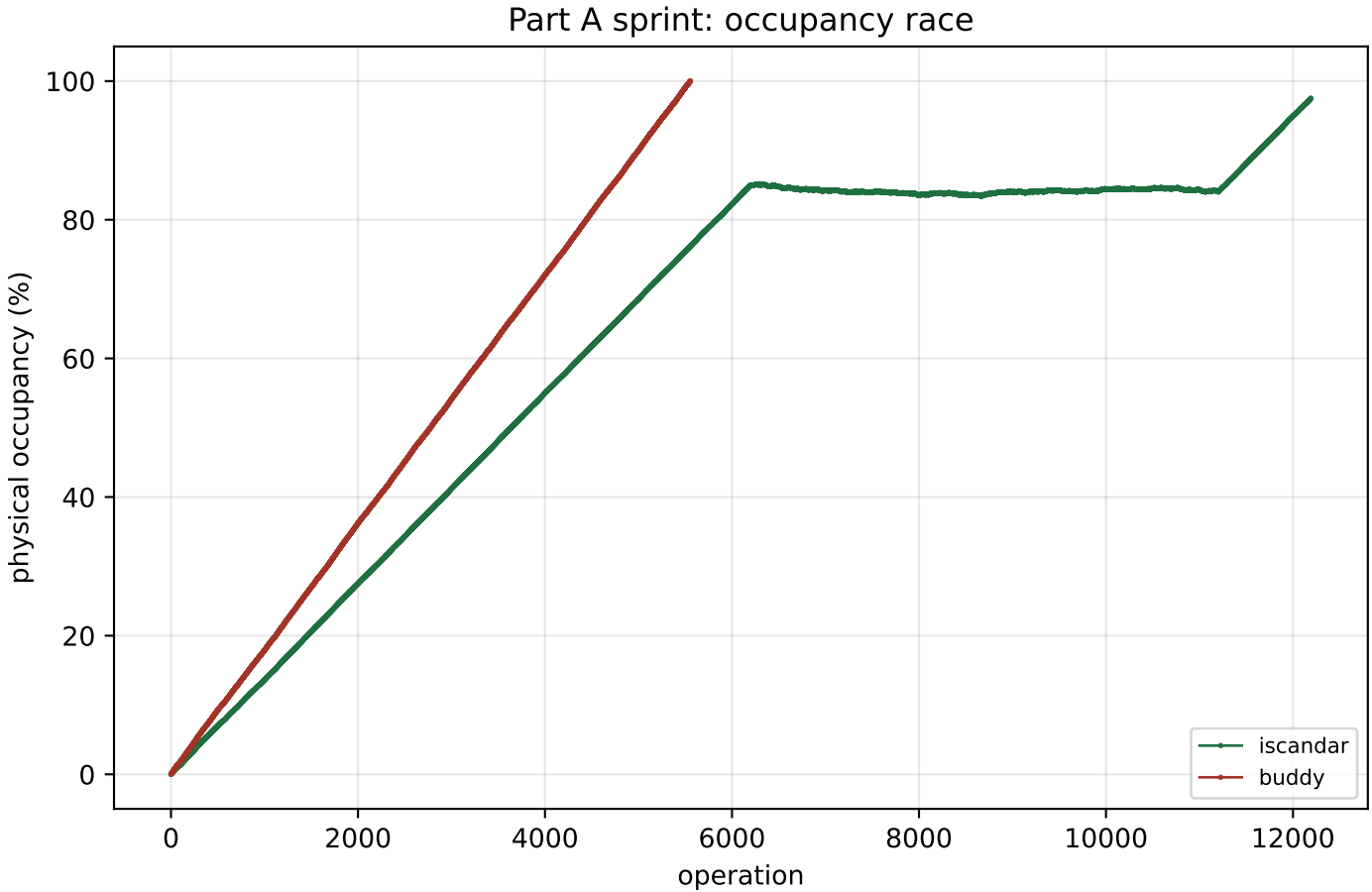
In the hold, Iscandar's allocation cost has median 14, with the middle half between 6 and 27 and the 90th percentile at 40. Buddy's has median 2, middle half 2 to 2, 90th percentile 2. The stability sigma on the closing page states the same shapes as a single number per series.


Largest free run over time (hold + push): fragmentation made visible
iscandar
buddy
largest free run (sectors)
operation
0
5000
10000
15000
20000
25000
30000

Commentary - largest free run

The fragmentation picture. Through the hold, Iscandar's largest contiguous free run has median 18976 sectors; Buddy's largest free block has median 8192. At the push stalls, Buddy dies with a largest free block of 1 sectors at 100.00% physical, Iscandar with a largest free run of 4 at 97.73% physical.

Graph 5 is the one that makes the RTOS applications engineer stop and look. The occupancy graphs show what Buddy wastes; the largest-free-run graph shows what that waste costs - the size of the biggest promise each allocator can still keep.


Part A sprint: occupancy race
physical occupancy (%)
operation
0
20
40
60
80
100
0
2000
4000
6000
8000
10000
12000
iscandar
buddy

Commentary - the sprint

buddy stalls first, at op 5555 (100.00% physical); iscandar continues alone and stalls at op 12191 (97.51% physical). The margin is 6636 ops to iscandar.

Numbers and caveats

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Numbers (population statistics over successful ops)

sprint iscandar alloc: n=9647 mean 8.40 var 170.38 CV 155.5% stability 0.6 sigma, range 3..73

sprint buddy alloc: n=5555 mean 4.00 var 8.00 CV 70.7% stability 1.4 sigma, range 2..26

hold   iscandar alloc: n=9988 mean 18.16 var 209.86 CV 79.8% stability 1.3 sigma, range 3..72
hold   iscandar free:  n=10012 mean 2.00 var 0.00 CV 0.0% stability inf sigma, range 2..2

hold   buddy alloc: n=9988 mean 2.04 var 0.08 CV 14.1% stability 7.1 sigma, range 2..10
hold   buddy free:  n=10012 mean 2.02 var 0.02 CV 7.4% stability 13.5 sigma, range 2..5

Counting and caveats

- Iscandar accesses are sector-word touches on
  the engine path: probe words in the search plus
  the commit's word touches, per the unified
  model in gaussian_race.py's header. Gate
  (counter) reads are a separate series and are
  not blended into accesses.
- Buddy accesses are its tree-work analog units
  (buddy.py), NOT the same physical units. Read
  across engines by shape, CV, and outcome;
  read levels and sigmas only within one engine
  against its own bound (the two-series
  discipline adopted 2026-10-26).
- Stability sigma is mean / SD (the reciprocal
  of CV): how many of its own standard
  deviations the mean cost sits above zero. It
  is a stability ratio, not a defect-rate
  sigma; a defect sigma needs a declared limit.
- Placement completeness (post-amendment): an
  Iscandar failure means no contiguous free run
  of the request existed, or the request
  exceeded all free space. It never fails facing
  an exact-fit hole.
- One seeded stream drove both engines in each
  part (seed 1). GammaCheck green throughout.
```